

Education

Purdue University - Main Campus

2025 - 2028

Cybersecurity B.S

GPA: 4.0

John Martinson Honors College, Purdue Presidential Scholarship Recipient

Polytechnic Student Council Member

John Martinson Honors College Leadership Committee Member

Achievements

- Purdue Design and Innovation Challenge 1st Place 2025
- Purdue BoilerSTORM Defense Competition 1st Place 2025
- Purdue University Hello World Hackathon 2nd Place 2025
- RU Secure CTF 1st Place 2025
- Lockheed Martin CyberQuest CTF 1st Place 2025
- Garden State CTF Tied for 1st Place 2025
- HackFree Revolutionizing Scholarship Research Award 2025
- RU Secure Nation Finalist 2024 - 2025
- Cyber Patriot 1st in State 2024
- CyberPatriot 4x ranked Platinum 2021 - 2024
- 2x National Cyber Scholarship Foundation Scholar with Honors 2023 - 2024
- 3x Cyberstart Gold Medalist 2021 - 2024
- HackFree Hack of Distinction 2023 - 2024

Certifications

- GIAC Security Essentials (GSEC) 2024
- IBM Cloud Essentials 2024
- Google Cybersecurity 2024
- Cisco Ethical Hacker 2024
- GIAC Foundational Cybersecurity Technologies(GFACT) 2023

Skills

- **Cybersecurity:** Network Security, Digital Forensics, OSINT, Cryptography, Reverse Engineering, Hardware Security
- **Programming:** Python, Java, JavaScript, SQL, React, Next.js, HTML/CSS
- **Cloud Technologies:** AWS, Azure, GCP, Docker, Kubernetes

Work Experience

- **Purdue Lab for Research in Emerging Network & Distributed Systems - Undergraduate Research Assistant** **September 2025 - Present**
 - Reverse engineering ARM firmware binaries using static disassembly and dynamic runtime analysis to identify memory corruption vulnerabilities, sensor spoofing attack vectors, etc. on ArduPilot based autonomous drones.
- **The Data Mine - Purdue University - Undergraduate Data Science Researcher** **August 2025 – Present**
 - Developing a malicious DNS domain detection pipeline for Elevance Health, aggregating DNS logs via Splunk and Infoblox, engineering behavioral and pattern features to detect DGA domains, and deploying an LSTM, Gradient Boosting, Isolation Forest, and DBSCAN models with a committee voting system to identify threats.
- **New Jersey Cybersecurity and Communications Integration Cell - Advanced Internship** **July 2025 - Aug 2025**
 - Led a cybersecurity project developing an AI-based system to detect deepfake images and videos. Helped design and train a convolutional neural network (CNN) to identify facial inconsistencies, lighting mismatches, and pixel-level artifacts commonly found in manipulated media.

Projects

- **Akawa | HackIllinois 2026**
 - Built a real-time security threat detection platform for existing CCTV infrastructure, integrating weapon detection (YOLOv11), violence detection (Two-Stream CNN-LSTM), and audio classification (LSTM on mel-spectrograms) with an automated incident pipeline triggering PDF generation, cloud storage, and email alerts within seconds
- **Pedal | Hello World Hackathon**
 - Built a safety-aware bike/pedestrian navigation app processing 350K+ nodes across Indianapolis with <500ms response times, using custom Dijkstra's with KDE-derived crime density surfaces and R-Tree spatial indexing to route users away from high-crime areas onto bike-lane-prioritized paths

Leadership

- **IngeniumSTEM Founder/President** **February 2024 - Present**
 - Founded and led IngeniumSTEM organization, directing a board to create and host STEM competitions. Designed and executed major hackathons with 2,000+ total participants across events, securing in-kind prizes through sponsor negotiations. Established partnerships with 30+ companies.